

Fourier Transforms Primer

[These notes rely on complex variables]

Fourier's idea:

we can write functions as superpositions of other (special) functions.

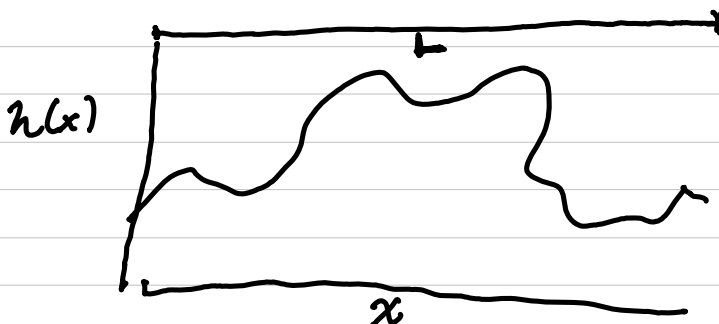
Fourier series:

periodic function $\rightarrow$ discrete sum of oscillating functions (complex exponential or sines and cosines)

Fourier transform:

periodic or non-periodic function $\rightarrow$ integral of complex exponentials

Consider some function



The "Fourier transform" of $h(x)$ is:

$$h(q) = \frac{1}{\sqrt{L}} \int_0^L dx h(x) e^{iqx}$$

the "inverse Fourier transform":

$$h(x) = \frac{1}{\sqrt{L}} \sum_q h(q) e^{-iqx}$$

$$q = \text{wave number} = \frac{2\pi}{\lambda}$$

2 wavelength

Example

$$h(x) = \cos(\omega_1 x) \text{ for } L=1$$

$$h(q) = \frac{1}{\sqrt{1}} \int_0^1 dx \cos \omega_1 x e^{iqx}$$

$$= \frac{-iq + \frac{e^{iq} (iq \cos \omega_1 - \omega_1 \sin \omega_1)}{q^2 - \omega_1^2}}{q^2 - \omega_1^2}$$

$$= \frac{-iq + [\underbrace{\cos q}_{\cos q} + i \underbrace{\sin(-q)}_{-\sin q}][iq \cos \omega_1 - \omega_1 \sin \omega_1]}{q^2 - \omega_1^2}$$

$$= \frac{1}{q^2 - \omega_1^2} \left[-\omega_1 \cos q \sin \omega_1 + q \sin q \cos \omega_1 + i(-q + q \cos q \cos \omega_1 + \omega_1 \sin q \sin \omega_1) \right]$$

$$= \frac{1}{q^2 - \omega_1^2} \left[-\omega_1 \cos q \sin \omega_1 + q \sin q \cos \omega_1 + i(-q + q \cos q \cos \omega_1 + \omega_1 \sin q \sin \omega_1) \right]$$

So taking the real and imaginary parts:

$$\text{Re}[h(q)] = \frac{-\omega_1 \cos q \sin \omega_1 + q \sin q \cos \omega_1}{q^2 - \omega_1^2}$$

$$\text{Im}[h(q)] = \frac{-q + q \cos q \cos \omega_1 + \omega_1 \sin q \sin \omega_1}{q^2 - \omega_1^2}$$

Define

$$f(q, \omega_1) = \omega_1 \cos q \sin \omega_1 + q \sin q \cos \omega_1$$

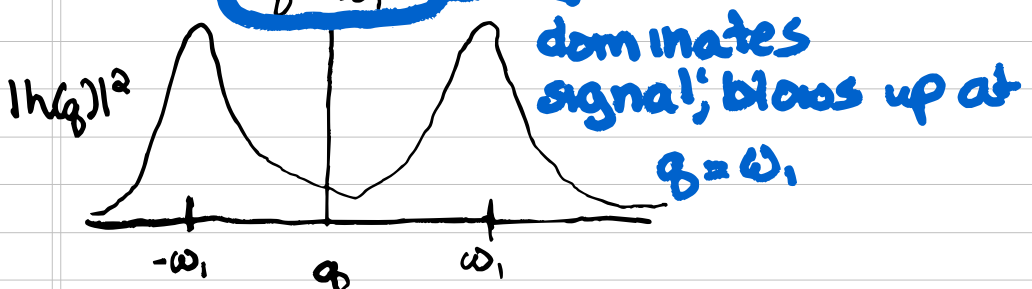
$$g(q, \omega_1) = q + q \cos q \cos \omega_1 + \omega_1 \sin q \sin \omega_1$$

$$\text{Re}[h(q)] = \frac{f(q, \omega_1)}{q^2 - \omega_1^2} \quad \text{Im}[h(q)] = \frac{g(q, \omega_1)}{q^2 - \omega_1^2}$$

The "spectrum" is given by

$$|h(q)|^2 = h(q)h^*(q) = [\text{Re}(h(q))]^2 + [\text{Im}(h(q))]^2$$

$$= \frac{f^2(q, \omega_1) + g^2(q, \omega_1)}{q^2 - \omega_1^2}$$



Example 2: Multiple frequencies

Suppose $L=1$ again and

$$\lambda(x) = a_1 \cos(\omega_1 x) + a_2 \cos(\omega_2 x)$$

then

$$h(q) = a_1 \int_0^1 \cos(\omega_1 x) e^{iqx} dx + a_2 \int_0^1 \cos(\omega_2 x) e^{iqx} dx$$

On the previous page we defined $f(q, \omega)$ and $g(q, \omega)$. Here, for short,

$$f_1 \equiv f(q, \omega_1); f_2 \equiv f(q, \omega_2) \quad | \quad g_1 \equiv g(q, \omega_1); g_2 \equiv g(q, \omega_2)$$

so

$$h(q) \approx \frac{a_1 (f_1 + i g_1)}{q^2 - \omega_1^2} + \frac{a_2 (f_2 + i g_2)}{q^2 - \omega_2^2}$$

$$\operatorname{Re}[h(q)] = \frac{a_1 f_1}{q^2 - \omega_1^2} + \frac{a_2 f_2}{q^2 - \omega_2^2} \quad | \quad \operatorname{Im}[h(q)] = \frac{a_1 g_1}{q^2 - \omega_1^2} + \frac{a_2 g_2}{q^2 - \omega_2^2}$$

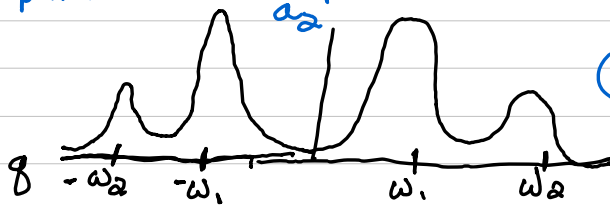
$$|h(q)|^2 = \operatorname{Re}^2 + \operatorname{Im}^2 =$$

$$= \frac{a_1^2 (f_1^2 + g_1^2)}{(q^2 - \omega_1^2)^2} + \frac{a_2^2 (f_2^2 + g_2^2)}{(q^2 - \omega_2^2)^2} + \frac{2a_1 a_2 (f_1 f_2 + g_1 g_2)}{(q^2 - \omega_1^2)(q^2 - \omega_2^2)}$$

peak at
 $q = \pm \omega_1$
with
amplitude
 a_1

peak at
 $q = \pm \omega_2$
with
amplitude
 a_2

cross term,
diverges more slowly
at $q = \omega_1$ or $q = \omega_2$

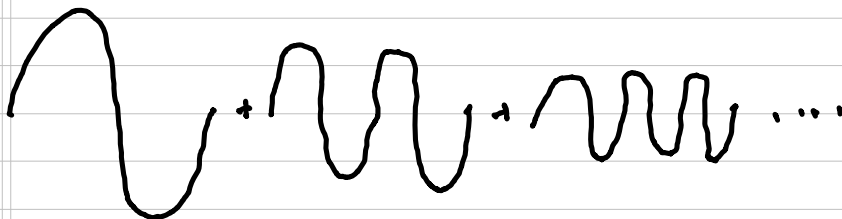


(assuming
 $a_1 > a_2$)

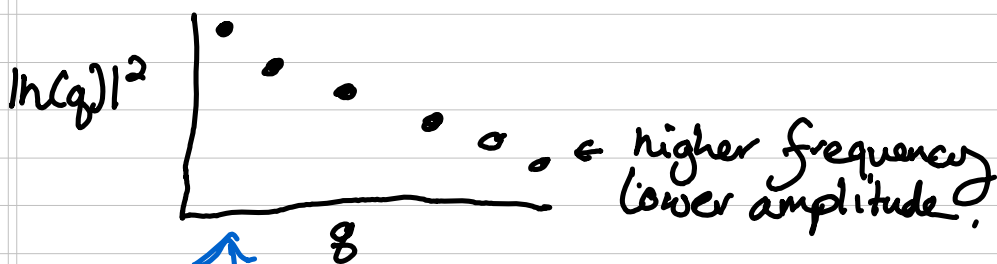
Example 3: Every possible g contributes
a mode

Imagine you had a lot of frequencies,
 $\omega_1, \dots, \omega_n$, and the amplitudes is
a function of the frequency, i.e.

$$a_i \propto \frac{1}{\omega_i} \leftarrow \begin{array}{l} \text{higher frequency,} \\ \text{lower amplitude} \end{array}$$



So the spectrum looks like



Also a spectrum!